
Prompt Augmentation in Medical Multimodal Large Language Models

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Abstract

Multimodal large language models (MLLMs) hold promise for clinical decision support, but tend to produce overconfident predictions that undermine their trustworthiness in safety-critical settings. We replicate Prompt4Trust, a reinforcement learning framework that trains a lightweight Calibration Guidance Prompt (CGP) Generator via Group Relative Policy Optimization (GRPO) to improve confidence calibration in a frozen downstream MLLM. We first verify replication fidelity on the PMC-VQA benchmark, then evaluate zero-shot generalization of the pretrained CGP Generator to an oral lesion classification dataset consisting of 567 intra-oral photographs across 13 lesion classes. Baseline evaluation reveals severe overconfidence across all three downstream MLLMs evaluated, with mean verbalized confidence of 91–96% against accuracies of 54–59%. Zero-shot CGP augmentation degrades accuracy on every model tested (−7.1 to −21.5 pp), and we identify reward hacking under domain shift as the root cause: the CGP aggressively narrows to specific answer candidates in 69% of oral lesion hints, but this narrowing carries no predictive signal.

1 Introduction

Multimodal large language models (MLLMs) are increasingly explored as tools for clinical decision support, offering the ability to reason over medical images and provide interpretable, interactive responses [1, 2]. However, two significant challenges hinder their deployment in safety-critical settings. First, MLLM output quality is highly sensitive to prompt design [3]. Second, MLLMs tend to hallucinate and express incorrect predictions with high confidence [4], a particularly dangerous property in clinical contexts where clinicians may rely on a model’s stated confidence to guide decisions.

Confidence calibration, the alignment between a model’s expressed confidence and its empirical accuracy, is therefore essential for trustworthy clinical deployment. Existing approaches such as verbalized self-assessment [5], sample consistency [6], and reflective prompting [3] attempt to address overconfidence but rely on manual prompt engineering and do not directly optimize calibration objectives. Prompt4Trust [7] introduced the first RL framework for prompt augmentation targeting confidence calibration, training a lightweight CGP Generator via GRPO [8] to improve verbalized confidence calibration in a frozen downstream MLLM.

Despite strong results on PMC-VQA [9], it remains unclear whether the framework generalizes beyond this radiology (the dominant category in PMC-VQA) or whether its instance-level reward produces reliable probability estimates under distributional shift. Importantly, it is unclear if the CGP generator shows reward hacking behaviour such as narrowing of choice from 4 to 2 or three most probable ones. In this work, we (1) replicate Prompt4Trust on PMC-VQA to verify reported results, (2) evaluate zero-shot generalization of the pretrained CGP Generator to oral lesion classification – a clinically distinct domain of 567 intra-oral photographs across 13 lesion classes (Appendix), and (3) conduct a systematic analysis of the CGP’s failure mode under domain shift, identifying reward hacking as the root cause.

2 Background

2.1 Confidence Calibration

A model is said to be *calibrated* if its expressed confidence aligns with its empirical accuracy: among all predictions made with confidence $\hat{p}$, approximately a fraction $\hat{p}$ should be correct. We measure calibration using two primary metrics. **Expected Calibration Error (ECE)** [10] partitions predictions into confidence bins and computes the weighted average of the absolute difference between mean confidence and mean accuracy within each bin. **Brier score** [11] measures the mean squared error between the predicted probability and the binary correctness outcome. We additionally report **calibration-in-large** (mean confidence minus accuracy) and **calibration slope** (logistic regression of correctness on logit-confidence, ideal = 1.0).

2.2 Verbalized Confidence

Rather than extracting confidence from model logits, *verbalized confidence* [5] instructs an MLLM to explicitly state a numerical confidence score alongside its answer. While this produces human-interpretable outputs, verbalized confidence is known to be poorly calibrated out of the box, with models tending to express uniformly high confidence regardless of actual accuracy [5, 6]. Prompt4Trust operates entirely in the verbalized confidence setting, as it targets frozen downstream MLLMs whose internal probability distributions are not accessible.

2.3 Group Relative Policy Optimization (GRPO)

GRPO [8] estimates advantages group-relatively across G sampled completions, avoiding the need for a separate value function. This makes it computationally efficient compared to PPO and well-suited to text generation settings with sparse rewards.

2.4 Prompt4Trust

Prompt4Trust [7] frames confidence calibration as a prompt optimization problem. A lightweight CGP Generator f_π takes the textual components of a medical VQA query and produces a context-aware auxiliary prompt $c = f_\pi(t \oplus A)$, which is appended to the original query before being passed to a frozen downstream MLLM f_τ : $(\hat{y}, \hat{p}) = f_\tau(v, t \oplus A \oplus c)$.

The CGP Generator is trained via GRPO using a clinically-motivated asymmetric reward:

$$r = \begin{cases} \log \min\{1, \max\{\hat{p}, \varepsilon\}\} & \text{if } \hat{y} = y \\ \log \min\{1, \max\{1 - \hat{p}, \varepsilon\}\} - 1 & \text{if } \hat{y} \neq y \\ \log(\varepsilon_{\text{penalty}}) & \text{otherwise} \end{cases} \quad (1)$$

where $\varepsilon = 10^{-10}$ ensures numerical stability and $\varepsilon_{\text{penalty}} = 10^{-12}$ penalizes invalid outputs.

3 Methodology

3.1 Replication on PMC-VQA

We replicate the Prompt4Trust framework following all reported hyperparameters. The CGP Generator is Qwen2.5-1.5B-Instruct [12] fine-tuned via GRPO on 5,000 randomly sampled questions from the PMC-VQA training split [9], with Qwen2-VL-2B-Instruct [13] as the frozen downstream MLLM. We follow the reported configuration ($\beta = 0.04$, top- $k = 100$, temperature = 0.6, lr = 10^{-6}), sampling 8 candidate completions per prompt during GRPO training.

3.2 Oral Lesion Dataset

We evaluate zero-shot generalization to a dataset of 567 intra-oral photographs spanning 13 lesion classes and approximately 10 anatomical site labels [14, 15]. Each image was paired with two multiple-choice questions: one asking for the lesion type and one asking for the anatomical site, each with four options, yielding 1,134 QA pairs.

3.3 Baseline and CGP Evaluation

We evaluate three frozen downstream MLLMs in a zero-shot verbalized confidence setting: Qwen2.5-VL-7B-Instruct [16], InternVL3-14B-Instruct [17] (Both of them evaluated in the original paper), and MedGemma-27b-it [18]. Each model was prompted with the image, question, and answer choices, instructed to respond in the format “the answer is {LETTER} with confidence {CONFIDENCE}.” Responses are parsed via regex; unparseable outputs are treated as invalid; as in the original paper. For CGP evaluation, the pretrained CGP Generator (xingshen/prompt4trust-cgpgenerator-1.5B) is applied zero-shot without any fine-tuning on oral lesion data.

3.4 Domain-Shift Analysis

To understand the mechanism behind CGP failure, we audit the generated hints for candidate-narrowing behaviour. We classify each hint as exhibiting *language narrowing* (contains phrases such as “likely to be. . .”) or *strict narrowing* (language narrowing plus explicit mention of 1–3 specific answer letters), and measure whether the ground-truth answer falls within the narrowed candidate set. We compare these rates between PMC-VQA (in-distribution) and oral lesions (out-of-distribution).

4 Experimental Results

4.1 Baseline Evaluation

All three downstream MLLMs exhibit severe overconfidence on oral lesion VQA in the zero-shot setting (Figure 1). Mean verbalized confidence ranges from 91% (InternVL3-14B) to 96% (MedGemma-27b), with Qwen2.5-VL-7B at 94%, while accuracies range from 54–59%. Notably, MedGemma-27b, the largest model and the only one explicitly trained on medical data, performs worst on lesion classification (50.6%) and is the most overconfident, suggesting that medical domain pre-training does not confer calibration benefits on out-of-domain visual tasks. Interestingly models except MedGemma-27b showed lower accuracy for anatomical site prediction (theoretically a simpler task) (Table 1).

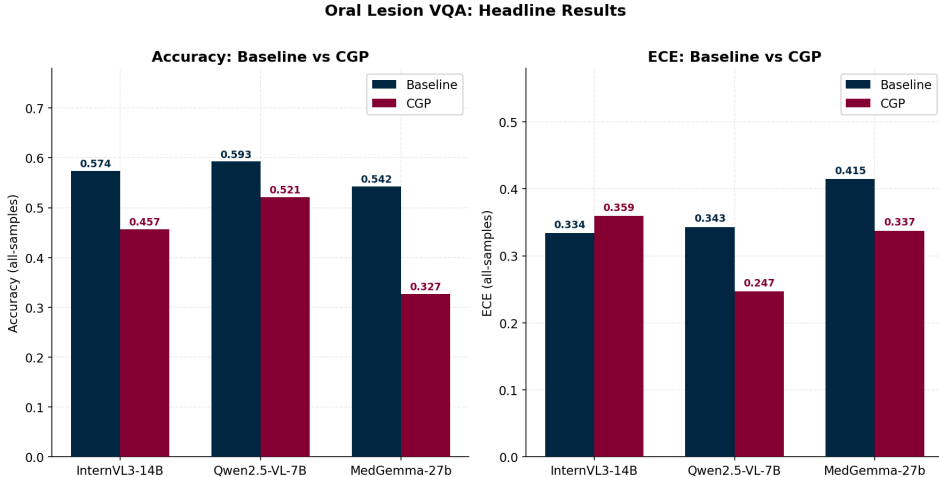


Figure 1: Accuracy and ECE for all three models, baseline vs. CGP augmentation.

Table 1: Baseline evaluation on oral lesion VQA.

Model	% valid response	Overall Accuracy	Lesion Acc.	Site Acc.	ECE
Qwen2.5-VL-7B	100%	59.3%	65.6%	52.9%	0.343
InternVL3-14B	100%	57.4%	62.1%	52.7%	0.334
MedGemma-27b	99.8%	54.2%	50.6%	57.8%	0.415

4.2 CGP Zero-Shot Generalization

Three architecturally distinct failure modes emerge. For Qwen2.5-VL-7B, accuracy drops modestly (−7.1 pp) but ECE improves substantially (−0.096), suggesting partial calibration benefit despite accuracy loss. For InternVL3-14B, both accuracy (−11.7 pp) and ECE (+0.026) worsen. For MedGemma-27b, CGP augmentation causes catastrophic format collapse on the lesion task - only 40.9% of lesion responses are parseable versus 99.6% at baseline - pulling overall accuracy down to 32.7%.

4.3 Domain-Shift Analysis: Reward Hacking

The CGP failure is explained by a distribution-dependent narrowing pathology (Figure 2) (Appendix A.2). On its training distribution (PMC-VQA), the CGP behaves correctly: it commits to a specific 1–3 letter candidate set in only 13.4% of hints, and when it does, the ground-truth answer is included 62.2% of the time - well above the 50% random baseline for size-2 candidate sets. On oral lesions, the CGP narrows aggressively in 69.1% of hints, but the ground-truth inclusion rate drops to 46.9%,

Table 2: CGP augmentation vs. baseline (all-samples, paper convention).

Model	Valid. (CGP)	Acc. base	Acc. CGP	Δ Acc	ECE base	Δ ECE
Qwen2.5-VL-7B	100%	0.593	0.521	-0.071	0.343	-0.096
InternVL3-14B	100%	0.574	0.457	-0.117	0.334	+0.026
MedGemma-27b	70.1%	0.542	0.327	-0.215	0.415	-0.077 [†]

[†] ECE improvement is confounded by 30% invalid responses counted as wrong.

indistinguishable from random. The site prediction task is worst: 55.4% strict narrowing with only 37.9% ground-truth coverage - 12.1 pp *below* random, meaning the CGP is actively misleading the downstream model.

Domain Shift: CGP Behaviour on PMC-VQA vs Oral Lesions

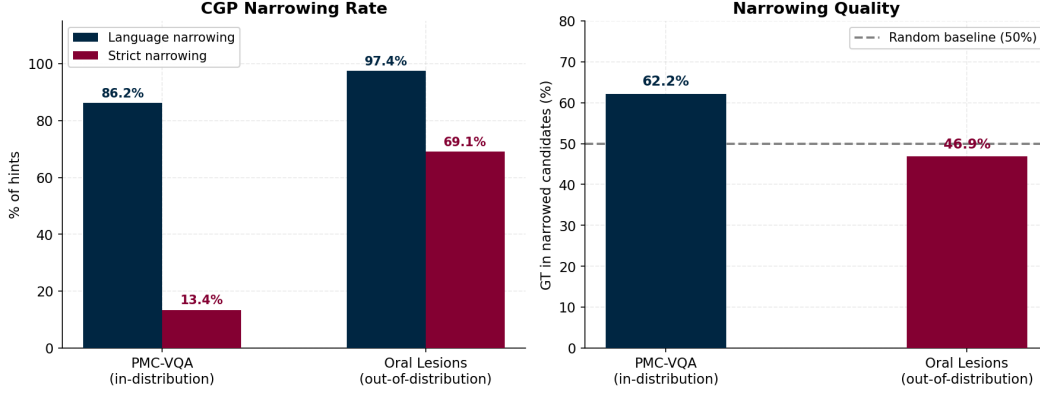


Figure 2: Domain shift analysis of CGP narrowing behaviour.

The Prompt4Trust reward function (Eq. 1) contains no neutrality term (e.g. penalization for including guesses of the answer in the calibration prompt generated by CGP), so the policy learned that narrowing hints maximise expected reward during training. Under domain shift, the learned policy continues to produce auxiliary prompts that narrow the options that the MLLM could choose. The per-class heatmap (Appendix Figure 5) confirms that CGP degradation is broad-spectrum across lesion classes, with no class consistently benefiting across all three models.

5 Conclusion and Future Work

We demonstrate that the pretrained CGP Generator degrades accuracy on all three downstream MLLMs when transferred zero-shot to oral lesion classification, and identify reward hacking as the root cause: the policy learned to narrow answer candidates during training and continues to do so OOD, where the narrowing carries no signal.

This finding highlights a fundamental limitation of instance-level calibration rewards without a neutrality constraint: they permit the policy to game the reward through confident-sounding but uninformative hints. The reward design in RL for NLP must account for OOD generalization, not just in-distribution performance. Specifically, in this case where the argument for post-training a smaller LLM as prompt modifier as CGP.

Future work should explore Reward Constrained Policy Optimization(RCPO)[19] for CGP generator, which formulates neutrality as a constraint $\mathbb{E}[\text{leakage}(c)] \leq \epsilon$ and uses Lagrangian dual ascent to enforce it asymptotically. A permutation-invariant reward, which scores hints across random letter-permutations of the answer choices, would also make narrowing strategically dominated.

6 Contributions and Acknowledgements

Ryan Rezai replicated Prompt4Trust on PMC-VQA and verified replication fidelity against reported baselines. Sreenath Madathil built the inference infrastructure, evaluation pipeline, and ran all model evaluations. Nusaibah Binte Rawnak conducted the domain-shift analysis and produced all figures and the final write-up. We used Claude Sonnet 4.6 (Anthropic) for conceptual clarification and code review during the project.

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A Oral lesion VQA Dataset

The oral lesions dataset consists of 567 intra-oral photographs extracted from open-access clinical case reports in dental and oral-pathology journals indexed on PubMed. Images, lesion-type labels, and anatomical-site labels were manually transcribed from the source publications. Two dentists independently reviewed image quality and label correctness - only images passing both reviews were retained. Each image carries two labels: a lesion type (13 classes - Aphthous ulcers, Squamous cell carcinoma, Mucocele, Gingival Hyperplasia, Hairy Tongue, Actinic Cheilitis Solar, Geographic Tongue, Gingivitis, Fordyce granules, Denture stomatitis, Leukedema, Cheek/lip/tongue chewing, Lymphoepithelial cyst, Gingival cyst) and an anatomical site (10 classes - Tongue, Gingiva, Inner-lip, Buccal-mucosa, Outer-lip, Floor-of-mouth, Palate, Retromolar trigone). This dataset is part of another benchmark data creation project, thus we are unable to provide the full data set for replication. A few example images are provided in the submission for reproducibility.

A.1 QA-pair construction

Each image is converted into two independent 4-choice MCQs (one for lesion type, one for site) for a total of 1134 QA pairs (src/generate_qa.py → data/orallesions/qa_pairs.csv). The procedure for each (image, task):

1. Question text is fixed natural language, identical across all images of that task — "What type of oral lesion is present in this intra-oral photograph?" and "What anatomical site is affected by the lesion in this intra-oral photograph?".
2. Options = the ground-truth class + 3 distractors sampled without replacement from the pool of all other class labels for that task. No LLM is involved in selecting distractors.
3. The four options are then shuffled and assigned letters A–D, giving a roughly uniform GT-letter distribution (observed: A 229, B 254, C 428, D 223 across both tasks combined).

The output schema (Figure_path, Question, Answer, Choice A–D, Task, Split) matches the PMC-VQA convention used by the original Prompt4Trust pipeline, so the same baseline-eval and CGP-eval scripts run on either dataset unchanged.

A.2 Example of remnants of reward hacking in domain-shift

A.2.1 Data point:

Oral lesion type QA Pair example - Ground truth: **Squamous cell carcinoma**

Question: What type of oral lesion is present in this intra-oral photograph?



Options:

- A. Actinic cheilitis solar
- B. Geographic Tongue
- C. Leukedema
- D. Squamous cell carcinoma

A.2.2 CGP-Generator Prompt

Prompt for CGP generator

System prompt:

You are assisting a downstream vision-language model (VLM) with answering a multiple-choice question about an image. You do not have access to the image or the correct answer. Do not give an answer to the question. Your task is to write a neutral hint based only on the question and answer options, to guide the VLM on what visual evidence to consider and what reasoning steps to use when answering.

User message template:

< Question >

Options:

A: <Choice A text>

B: <Choice B text>

C: <Choice C text>

D: <Choice D text>

Prompt draft:

A.2.3 Example of auxiliary prompt from CGP generator

The auxiliary prompt shows choice option narrowing from four to two choices. In this case, the two options are wrong (bolded text). The ground truth label is **Option D. Squamous cell carcinoma**

Prompt generated by CGP for above data point

The correct answer is likely to be either A: Actinic cheilitis or B: Geographic Tongue. However, without seeing the actual image, it's difficult to determine which specific type of lesion is present. The image could show any of these conditions, or something else entirely. To provide a neutral hint, you should focus on the general appearance and characteristics of the lesion, such as its location, size, and any notable features that might suggest a particular condition.

A.2.4 Prompt for MLLM

Prompt generated by CGP for above data point

You are answering a multiple-choice question with four options (A, B, C, or D). Clearly state your final answer and confidence in the following format only:

'In conclusion, the answer is LETTER with confidence CONFIDENCE.'

Replace LETTER with one of A, B, C, or D, and CONFIDENCE with a number between 0 and 100. After giving the answer, explain your reasoning based on the image and the question.

<Question>

<Image>

<Options>

<CGP Generated auxiliary prompt>

Answer:

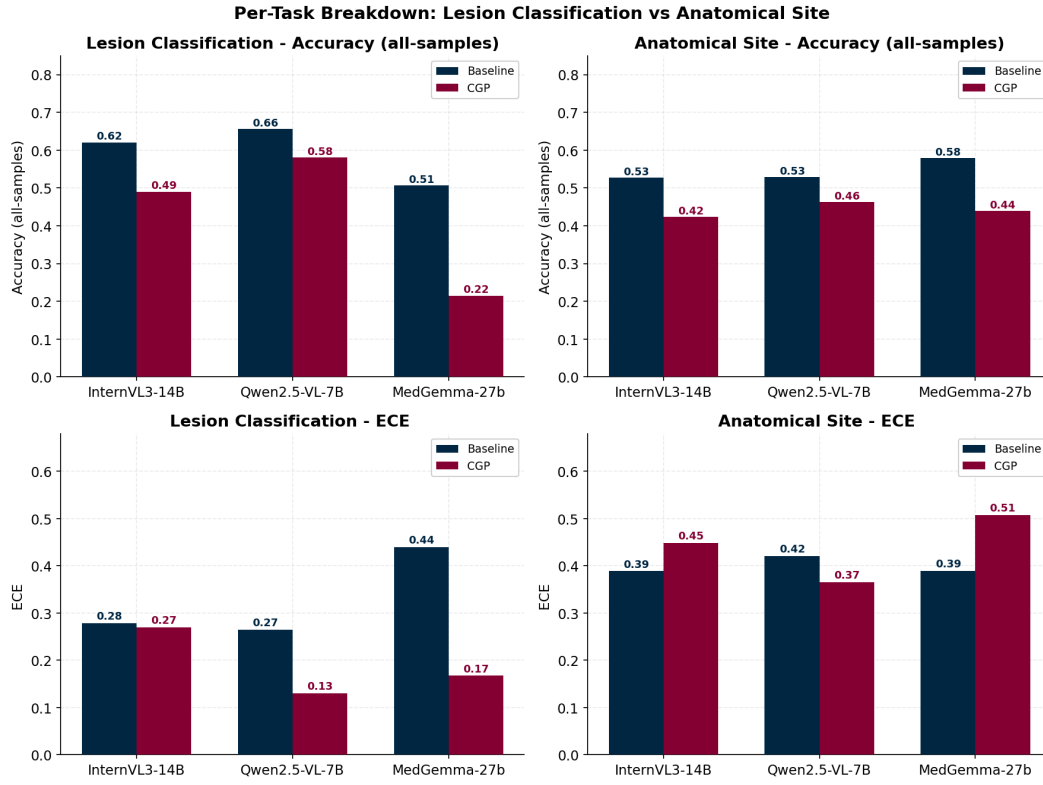


Figure 3: Per-task breakdown of accuracy and ECE for lesion classification vs. anatomical site prediction, baseline vs. CGP.

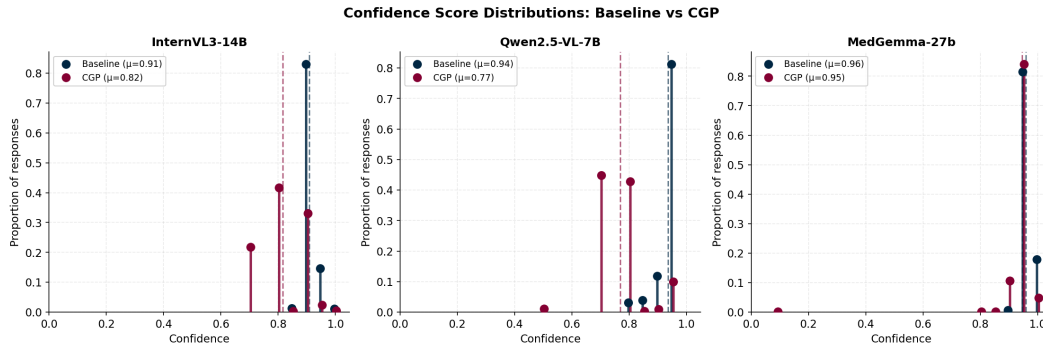


Figure 4: Confidence score distributions for all three models, baseline vs. CGP. All models are heavily concentrated at high confidence values.



Figure 5: Per-class accuracy change (CGP – Baseline) for the lesion classification task. Pink/red = accuracy drop, green = improvement.

B Additional Figures

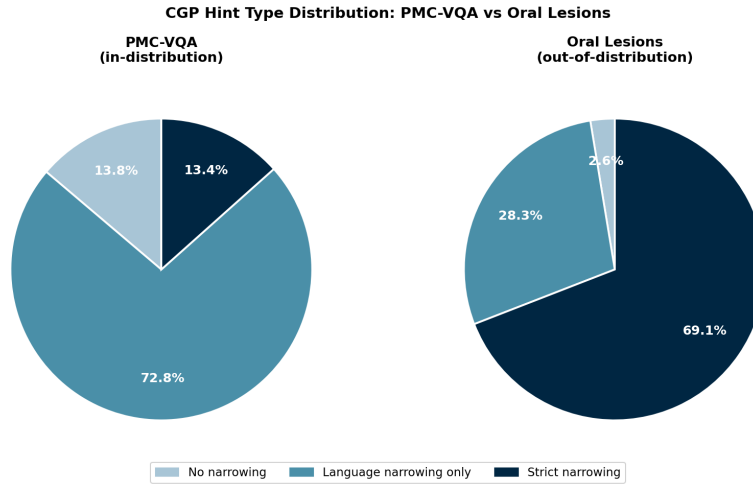


Figure 6: CGP hint type distribution on PMC-VQA vs. oral lesions. Strict narrowing dominates OOD hints despite carrying no predictive signal.

C Additional Tables

Table 3: Per-task accuracy and ECE, all-samples.

Model	Task	Acc. base	Acc. CGP	ECE base	ECE CGP
Qwen2.5-VL-7B	Lesion	0.656	0.580	0.265	0.131
	Site	0.529	0.462	0.421	0.365
InternVL3-14B	Lesion	0.621	0.490	0.279	0.270
	Site	0.527	0.423	0.389	0.449
MedGemma-27b	Lesion	0.506	0.215	0.440	0.168
	Site	0.579	0.439	0.389	0.507

Table 4: Valid-only metrics (selection-biased for MedGemma-27b where CGP validity = 70.1%).

Model	Acc. base	Acc. CGP	Δ Acc	ECE base	ECE CGP	Δ ECE	Brier base	Brier CGP
Qwen2.5-VL-7B	0.593	0.521	-0.071	0.343	0.247	-0.096	0.361	0.316
InternVL3-14B	0.574	0.457	-0.117	0.334	0.359	+0.026	0.352	0.375
MedGemma-27b	0.543	0.467	-0.077	0.415	0.481	+0.066	0.417	0.479

Note: MedGemma valid-only metrics exclude 30% unparsed responses and are selection-biased.

Table 5: Calibration metrics - baseline vs. CGP (valid-only responses).

Model	Cal.-in-large base	Cal.-in-large CGP	Δ	Slope base	Slope CGP	Δ
Qwen2.5-VL-7B	0.343	0.247	-0.096	-0.157	0.099	+0.257
InternVL3-14B	0.334	0.359	+0.026	1.441	0.254	-1.187
MedGemma-27b	0.415	0.479	+0.064	0.128	0.039	-0.089

Calibration-in-large = mean confidence - accuracy; ideal = 0. Slope ideal = 1.0.

Table 6: Mean confidence on incorrect predictions - baseline vs. CGP.

Model	Mean conf. base	Mean conf. CGP	Conf. incorrect base	Conf. incorrect CGP
Qwen2.5-VL-7B	0.936	0.768	0.937	0.767
InternVL3-14B	0.908	0.816	0.903	0.809
MedGemma-27b	0.959	0.946	0.950	0.525 [†]

[†] MedGemma confidence on incorrect is low due to format collapse. Invalid responses assigned $\varepsilon_{\text{penalty}}$.